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Complementary step and tree engaging structure for a climbing stick

Abstract

A climbing system includes a treestand and a plurality of climbing sticks. The treestand includes a monolithic platform formed from strengthened material. Each of the climbing sticks includes a frame having a plurality of weight-reduction apertures formed therethrough. Tree engaging structures include ramped surfaces to facilitate smooth downward sliding on a tree. Methods of manufacturing a treestand and climbing sticks are also disclosed. The method of manufacturing a treestand includes providing a solid, strengthened piece of material and removing portions of the material to form openings between structural supports. The method of manufacturing a climbing stick includes providing a frame and forming a plurality of weight-reduction apertures therethrough.

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Background/Summary

RELATED APPLICATIONS (1) This application is a divisional of co-pending U.S. patent application Ser. No. 16/943,830, filed Jul. 30, 2020, by the same inventors which is a continuation-in-part of co-pending U.S. patent application Ser. No. 16/141,579, filed Sep. 25, 2018, by the same inventors which is a divisional of co-pending U.S. patent application Ser. No. 15/997,463 filed Jun. 4, 2018, by the same inventors, which claims the benefit of U.S. Provisional Patent Application No. 62/548,384, filed on Aug. 21, 2017, by the same inventors, and also claims the benefit of U.S. Provisional Patent Application No. 62/622,960, filed on Jan. 28, 2018, by the same inventors, all of which are incorporated herein by reference in their respective entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This invention relates generally to elevated hunting platforms, and more particularly to portable treestand systems.

Description of the Background Art

(2) Portable treestand systems have become commonplace in big game hunting. Typically, a portable treestand system includes a treestand and a plurality of climbing sticks. The treestand includes a platform, a frame, a seat, and a means for securing the treestand to a tree. The platform provides a flat base for the hunter to stand on while the seat provides a place for the hunter to sit. Typically, the platform and the seat are mounted on the bottom and top ends, respectively, of the frame. The securing means is typically some type of suitable fastener(s) (e.g., ratchet strap, chain, steel cable, rope, etc.) that secures the frame to the tree. Each climbing stick typically includes a set of steps, a set of tree engaging features, a frame, and a means (e.g., ratchet strap, chain, steel cable, rope, etc.) for securing the climbing stick to the tree. The steps are affixed to the frame to provide surfaces to grab and step on when ascending and descending the tree. The tree engaging features are also affixed to the frame to facilitate the mounting of the climbing stick to the tree.

(3) In designing and manufacturing treestand systems, there are several factors that must be taken into consideration. For example, mobile hunters typically carry their treestand systems to their hunting locations, and it is, therefore, desirable to make the system as lightweight and compact as possible. Of course, in making the system as lightweight and compact as possible, it must also maintain a structural integrity sufficient for safe use. As another example, it is essential that the system be extremely quiet both when being mounted to a tree and also during the hunt so as not to alert game of the hunter's presence.

(4) There are several prior art treestand system designs that attempt to minimize weight. For example, treestand platforms are often formed by welding metal mesh onto square stock that has been bent into the shape of the platform. Likewise, the climbing sticks are often formed from lightweight metal tube having steps and tree engaging features welded thereon. Although such an approach can provide for a lightweight treestand system, it also suffers from undesirable noise caused by flexing materials and poor welds. In attempting to address problems associated with flexing materials and poor welds, some manufacturers have developed treestands having cast metal platforms that minimize weld points. Although cast metal platforms minimize weld points, they suffer other drawbacks. For example, cast metals are inherently brittle and, therefore, the platform has to be relatively large to compensate for the undesirable mechanical properties of cast metal. Of course, the added size results in a relatively heavy and bulky treestand. In addition, the failure of brittle materials tend to be catastrophic (e.g., shattering instead of bending), therefore causing significant safety problems.

(5) Another problem with known treestands and climbing sticks is that unexpected downward shifts of position on a tree can cause a climber to experience momentary fright, loss of balance, or even a fall from the tree. Downward shifts of even a few inches can cause such undesirable experiences.

(6) What is needed, therefore, is a treestand system that is lighter, stronger, smaller, more silent, and/or less prone to catastrophic failure than prior treestand systems. What is also needed is a treestand system that is more stable when attached to a tree.

SUMMARY

(7) The present invention overcomes the problems associated with the prior art by providing a portable treestand and climbing stick system that is lighter, safer, and more versatile than prior art systems.

(8) An example method of manufacturing a treestand includes providing a seat, providing one or more tree engaging structures, and providing a monolithic platform. The monolithic platform is formed by removing portions of a solid piece of material to form openings between a top surface of the platform and a bottom surface of the platform and to form structural supports between the openings. The seat, the one or more tree engaging structures, and the monolithic platform are combined to create a treestand.

(9) In a specific example, the method further includes subjecting the solid piece of material to a strengthening process. In a more specific example, the solid piece of material is subjected to the material strengthening process before the step of forming openings between the top surface of the platform and the bottom surface of the platform. In an even more specific example, the solid piece of material includes aluminum.

(10) In another specific example, the step of removing portions of the solid piece of material includes cutting the solid piece of material using a computer numerical control (CNC) system. In a more specific example, the step of removing portions of the solid piece of material includes machining the solid piece of material using a computer numerical control (CNC) system.

(11) In one specific example, the step of removing portions of the solid piece of material includes using an automated system to remove portions of the solid piece of material.

(12) In yet another example, the step of providing one or more tree engaging structures includes removing portions of the solid piece of material to form at least one of the tree engaging structures integral with the platform as a single monolithic structure. In a more specific method, an intermediate portion of the monolithic structure between the platform and the tree engaging structure has a width, and the width of the intermediate portion of the monolithic structure is narrower than a width of the platform. The width of the intermediate portion of the monolithic structure can also be narrower than a width of the integral tree engaging structure. In an even more specific example, the method further includes providing a tubular frame structure and coupling the frame structure to the intermediate portion of the monolithic structure. The tubular frame structure has a length and a cross-sectional dimension. The width of the intermediate portion of the

monolithic structure is no more than twice the cross-sectional dimension of the tubular frame structure.

(13) Another example method further includes providing a tubular frame structure including a top end, a bottom end, and a plurality of weight reduction apertures arranged between the top end and the bottom end. The example method further includes coupling the seat to the top region of the frame structure and coupling the platform to the bottom region of the frame structure. In a more specific example, the step of coupling the platform to the bottom region of the frame structure includes providing a hinge bracket, fixably mounting the hinge bracket on the platform, and hingably coupling the frame structure to the hinge bracket.

(14) An even more specific example further includes providing a first tensile support, a first fastener, a second tensile support, and a second fastener. The first tensile support includes a first end and a second end, and the second tensile support includes a first end and a second end. The example also includes forming a first aperture through a first peripheral sidewall of the platform. The first aperture is disposed below a top surface of the structural supports of the platform and above a bottom surface of the structural supports of the platform. The example also includes coupling the first end of the first tensile support to the first aperture with the first fastener and coupling the second end of the first tensile support to the frame structure. The example also includes forming a second aperture through a second peripheral sidewall of the platform. The second aperture is disposed below the top surface of the structural supports of the platform and above the bottom surface of the structural supports of the platform. The example also includes coupling the first end of the second tensile support to the second aperture with the second fastener and coupling the second end of the second tensile support to the frame structure.

(15) In yet another example method, the seat is a monolithic structure formed by removing portions of a solid piece of material to form openings between a top surface of the seat and a bottom surface of the seat, forming structural supports therebetween.

(16) Another more specific example method further includes removing a portion of the solid piece of material to form an integral bow receiving feature in the platform, providing a bow engaging device, and adjustably coupling the bow engaging device to the platform.

(17) In another specific example method, forming the monolithic platform includes forming at least some recesses in the top surface of the platform but leaving portions of the bottom surface of the platform between adjacent structural supports formed by the recesses.

(18) An example treestand is also described. The example treestand includes a seat, a first tree engaging structure, and a platform. The platform is coupled to the first tree engaging structure and is made of a strengthened material defining a web of structural supports that define openings between a top surface of the platform and a bottom surface of the platform.

(19) In a particular embodiment, the strengthened material includes strengthened metal. In a more particular embodiment, the strengthened metal includes aluminum. In an even more particular embodiment, the strengthened metal includes an aluminum alloy with magnesium and silicon. In another particular embodiment, the strengthened metal is an extruded metal.

(20) In an example embodiment, the tree engaging structure and the platform are integral parts of a monolithic structure. In a more particular embodiment, an intermediate portion of the monolithic structure between the platform and the tree engaging structure has a width, and the width of the intermediate portion of the monolithic structure is narrower than a width of the platform. The width of the intermediate portion of the monolithic structure is also narrower than a width of the tree engaging structure.

(21) In a more particular embodiment, the treestand further includes a tubular frame having a length and a cross-sectional dimension. The frame structure is coupled to the intermediate portion of the monolithic structure, and the width of the intermediate portion of the monolithic structure is no more than twice the cross-sectional dimension of the tubular frame.

(22) In another particular embodiment, the treestand further includes a tubular frame having a top

end and a bottom end. The seat is hingably coupled to the top end of the frame, and the platform is hingably coupled to the bottom end of the frame. In a more particular example, the treestand further includes a hinge bracket hingably coupled to the frame. The hinge bracket is formed separately from, and fixed to, the platform.

(23) In a more particular example, the treestand further includes a first tensile support, a second tensile support, a first fastener, a second fastener, a first aperture, and a second aperture. The first tensile support includes a first end and a second end. Likewise, the second tensile support includes a first end and a second end. The first aperture is disposed through a first peripheral sidewall of the platform, below a top surface of the structural supports of the platform, and above a bottom surface of the structural supports of the platform. The second aperture is disposed through a second peripheral sidewall of the platform, below the top surface of the structural supports of the platform, and above the bottom surface of the structural supports of the platform. The first end of the first tensile support is coupled to the first aperture by the first fastener, and the second end of the first tensile support is coupled to the frame structure. The first end of the second tensile support is coupled to the second aperture by the second fastener, and the second end of the second tensile support is coupled to the frame structure.

(24) In yet another example embodiment, the seat is a monolithic structure made of a solid piece of strengthened material and includes a web of support structures defining openings between a top surface of the seat and a bottom surface of the seat.

(25) Optionally, the treestand further includes an integral bow receiving feature defined by the platform and a bow engaging device adjustably coupled to the platform.

(26) In yet another particular embodiment, the platform includes at least some recesses formed in the top surface of the platform but leaving portions of the bottom surface of the platform between adjacent structural supports formed by the recesses.

(27) Example climbing sticks are also disclosed. An example climbing stick includes a frame tube, a first step, a second step, a first tree engaging structure, and a second tree engaging structure. The frame tube extends along an axis and includes a first end and a second end. The first step is mounted to the first end of the frame tube, and the second step is mounted to the second end of the frame tube. The first tree engaging structure is fixed to the first end of the frame tube, and the second tree engaging structure is fixed to the second end of the frame tube. The frame tube includes a plurality of weight reduction apertures formed therethrough.

(28) In an example embodiment, the frame tube includes a front wall, a first side wall, a rear wall, and a second side wall. The first side wall is disposed between the front wall and the rear wall. The second side wall is also disposed between the front wall and the rear wall, on an opposite side of the front and rear walls. The plurality of weight reduction apertures include a first plurality of apertures formed in the front wall of the frame tube. The plurality of weight reduction apertures additionally includes a second plurality of apertures formed in the first side wall of the frame tube. Apertures of the first plurality of apertures formed in the front wall of the frame tube are vertically offset with respect to adjacent apertures of the second plurality of apertures formed in the first side wall of the frame tube. In a more particular embodiment, the plurality of weight reduction apertures includes a third plurality of apertures passing through the rear wall of the frame tube. Apertures of the first plurality of apertures are vertically aligned with corresponding apertures of the third plurality of apertures. The plurality of weight reduction apertures includes a fourth plurality of apertures passing through the second side wall of the frame tube. Apertures of the second plurality of apertures are vertically aligned with corresponding apertures of the fourth plurality of apertures.

(29) In an even more particular embodiment, the climbing stick is configured to stack on a second climbing stick in a substantially horizontally aligned position, when the frame tube of the climbing stick and the frame tube of the second climbing stick are oriented horizontally. An example climbing stick is configured to engage a second climbing stick in an aligned position, with corresponding features of the climbing stick and the second climbing stick substantially aligned

along an axis perpendicular to the axis of the frame tube of the climbing stick and to a corresponding axis of a frame tube of the second climbing stick. In a more particular example, the first tree engaging structure includes a frame engaging feature configured to engage the frame tube of the second climbing stick. The first step includes a restraining feature facilitating the engagement of the frame tube by a tree engaging structure of a third climbing stick and limiting relative motion between the climbing stick and the third climbing stick. The second tree engaging structure includes a frame engaging feature configured to engage the frame tube of the second climbing stick. The second step includes a restraining feature facilitating the engagement of the frame tube by a second tree engaging structure of the third climbing stick and limiting relative motion between the climbing stick and the third climbing stick.

(30) In yet another particular example, a portion of each of the first and second steps is configured to receive a respective tree engaging structure of another climbing stick. In a more particular example, the portions of the steps each includes a restraining feature configured to prevent the tree engaging structures of the other climbing stick from sliding along the frame tube.

(31) In one particular embodiment, at least one of said tree engaging structures includes a first tooth and a second tooth. The first tooth and the second tooth are adapted to bite into a tree, and the first tooth and the second tooth are configured together to receive a frame tube of the other climbing stick therebetween.

(32) Optionally, at least one of the tree engaging structures defines a channel configured to seat the frame tube. The channel prevents rotation of the tree engaging structure(s) relative to the frame tube, when the tree engaging structure(s) is/are fixed to the frame tube.

(33) A method of manufacturing a climbing stick is also disclosed. The method includes providing a frame tube, a first step structure, a second step structure, a first tree engaging structure, and a second tree engaging structure. The frame tube includes a first end, a second end, and a plurality of weight reduction apertures between the first end and the second end. The method further includes mounting the first step structure to the first end of the frame tube, mounting the second step structure to the second end of the frame tube, mounting the first tree engaging structure to the first end of the frame tube, and mounting the second tree engaging structure to the second end of the frame tube.

(34) In a particular embodiment, the frame tube includes a front wall, a first side wall, a rear wall, and a second side wall. The first side wall is disposed between the front wall and the rear wall. The second side wall is disposed between the front wall and the rear wall, opposite the first side wall. The plurality of weight reduction apertures include a first plurality of apertures formed in the front wall of the frame tube. The plurality of weight reduction apertures additionally includes a second plurality of apertures formed in the first side wall of the frame tube. Apertures of the first plurality of apertures formed in the front wall of the frame tube are vertically offset with respect to adjacent apertures of the second plurality of apertures formed in the first side wall of the frame tube. In a more particular example, the plurality of weight reduction apertures includes a third plurality of apertures passing through the rear wall of the frame tube. Apertures of the first plurality of apertures are vertically aligned with corresponding apertures of the third plurality of apertures. In addition, the plurality of weight reduction apertures includes a fourth plurality of apertures passing through the second side wall of the frame tube, and apertures of the second plurality of apertures are vertically aligned with corresponding apertures of the fourth plurality of apertures.

(35) An example method further includes configuring the climbing stick to stack on a second climbing stick in a substantially horizontally aligned position, when the frame tube of the climbing stick and the frame tube of the second climbing stick are oriented horizontally. A more particular example method includes configuring the climbing stick to engage a second climbing stick in an aligned position, with corresponding features of the climbing stick and the second climbing stick substantially aligned along an axis perpendicular to the axis of the frame tube of the climbing stick and a corresponding axis of a frame tube of the second climbing stick. In an even more particular

example method, the first tree engaging structure includes a frame engaging feature configured to engage the frame tube of the second climbing stick. In addition, the first step includes a restraining feature facilitating the engagement of the frame tube by a tree engaging structure of a third climbing stick and limiting relative motion between the climbing stick and the third climbing stick. The second tree engaging structure includes a frame engaging feature configured to engage the frame tube of the second climbing stick. The second step includes a restraining feature facilitating the engagement of the frame tube by a second tree engaging structure of the third climbing stick and limiting relative motion between the climbing stick and the third climbing stick.

(36) In yet another particular method, a portion of each of the first step and the second step is configured to receive a respective tree engaging structure of another climbing stick. In a more particular example embodiment, the portions of the steps each include a restraining feature configured to prevent the tree engaging structures of another climbing stick from sliding along the frame tube.

(37) In another particular method, the tree engaging structure includes a first tooth and a second tooth. The first tooth and the second tooth are adapted to bite into a tree. In addition, the first tooth and the second tooth are configured together to receive a frame tube of another climbing stick therebetween.

(38) In another example method, the first tree engaging structure and the second tree engaging structure each defines a channel configured to seat the frame tube. The frame tube is seated in the channel, and the channel prevents rotation of the tree engaging structure relative to the frame tube, when the tree engaging structure is fixed to the frame tube.

(39) Example tree engaging structures that reduce erratic and unpredictable movement of tree mounted structures are also disclosed. One example tree mountable structure includes a frame, a first step, a first tree engaging structure, and a fastener. The first step is coupled to the frame and is configured to support a user standing thereon. The first tree engaging structure is coupled to the frame and includes a facing surface. The facing surface has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge. The fastener is configured to attach to the frame and to draw the facing surface against a tree. The top edge of the facing surface is disposed closer to a center of the tree than the bottom edge of the facing surface when the fastener draws the facing surface against the tree. The ramped surface facilitates the sliding of the first tree engaging structure down the tree, and the top edge of the facing surface of the first tree engaging structure resists the sliding of the first tree engaging structure up the tree.

(40) One particular example tree mountable structure is a climbing stick. The example climbing stick additionally includes a second tree engaging structure. The first tree engaging structure is attached to a first end of the frame, and the second tree engaging structure is attached to an opposite second end of the frame. The second tree engaging structure includes a facing surface that has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge of the second tree engaging structure. The fastener is configured to also draw the facing surface of the second tree engaging structure against the tree. The top edge of the facing surface of the second tree engaging structure is also disposed closer to the center of the tree than the bottom edge of the facing surface of the second tree engaging structure when the fastener draws the facing surface of the second tree engaging structure against the tree. The ramped surface of the second tree engaging structure facilitates the sliding of the second tree engaging structure down the tree, and the top edge of the facing surface of the second tree engaging structure resists the sliding of the second tree engaging structure up the tree. Optionally, the example climbing stick additionally includes a second step coupled to the frame and being configured to support the user standing thereon, the first step being coupled to the first end of the frame, and the second step being coupled to the second end of the frame.

(41) Another example tree mountable structure is a treestand. In an example treestand, the step is a platform configured to support a user standing thereon. Optionally, the first tree engaging structure

is an integral part of the platform.

(42) An example treestand additionally includes a second tree engaging structure. The first tree engaging structure is attached at a first end of the frame, and the second tree engaging structure is attached at an opposite second end of the frame. The second tree engaging structure also includes a facing surface that has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge of the second tree engaging structure. The fastener is configured to also draw the facing surface of the second tree engaging structure against the tree. The top edge of the facing surface of the second tree engaging structure is disposed closer to the center of the tree than the bottom edge of the facing surface of the second tree engaging structure when the fastener draws the facing surface of the second tree engaging structure against the tree. The ramped surface of the second tree engaging structure facilitates the sliding of the second tree engaging structure down the tree, and the top edge of the facing surface of the second tree engaging structure resists the sliding of the second tree engaging structure up the tree.

(43) Optionally, the first tree engaging structure and/or the second tree engaging structures are each monolithic structures. The first and/or second tree engaging structures can define a first channel configured to seat the frame. The channel prevents rotation of the tree engaging structures relative to the frame.

(44) In some example tree mountable structures, the ramped surface of the tree engaging structure is planar. In other example devices, the ramped surfaces are curved/rounded surfaces.

(45) In example tree engaging structures, the facing surface defines a plurality of teeth configured to penetrate the tree when the facing surface is urged against the tree. A first one of the teeth includes an apex extending vertically. A first portion of the facing surface and a second portion of the facing surface converge to form the apex of the first tooth. A second one of the teeth also includes an apex extending vertically. The first portion of the facing surface and a third portion of the facing surface converge to form the apex of the second tooth. The ramped surface extends between a lower end of the apex of the first tooth and the bottom edge of the tree engaging structure, and a second ramped surface extends between a lower end of the apex of the second tooth and the bottom edge of the tree engaging structure. Optionally, the first portion of the facing surface includes a concave surface extending between the apex of the first tooth and the apex of the second tooth.

(46) An example tree engaging structure includes a plurality of teeth, each of the teeth including an apex, a first side face, and a second side face. The apex of each tooth extends upwardly from a bottom point to a top point. The first side face converges toward the apex, and the first side face and the ramped surface share a first common edge. The first side face is bounded at least in-part by the top edge of the facing surface, the bottom edge of the facing surface, and the first common edge. The second side face also converges toward the apex, and the second side face and the ramped surface also share a second common edge. The second side face is bounded at least in-part by the top edge of the facing surface, the bottom edge of the facing surface, and the second common edge. The ramped surfaces face in a downward direction when the climbing stick is mounted to the tree, and the first common edge and the second common edge converge upwardly toward the bottom point of the apex.

(47) In example tree engaging structures, the first side face shares a third edge with a first portion of the top edge of the facing surface. The second side face also shares a fourth edge with a second portion of the top edge of the facing surface. The third edge and the fourth edge converge toward the top point of the apex. Optionally, the first side face includes a concave surface, and the second side face includes a concave surface.

(48) An example treestand includes a frame, a standing platform, a first tree engaging structure, a second tree engaging structure, and a fastener. The standing platform is coupled to the frame and is configured to support a user standing thereon. The first tree engaging structure is also coupled to the frame and includes a facing surface. The facing surface has a top edge, a bottom edge, and a

ramped surface extending between the top edge and the bottom edge. The second tree engaging structure is formed integrally with the platform and also includes a facing surface. The facing surface of the second tree engaging structure also has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge. The fastener is configured to attach to the frame and to draw the facing surfaces of the first tree engaging structure and the second tree engaging structure against a tree. The top edges of the facing surfaces of the first and second tree engaging structures are disposed closer to a center of the tree than the bottom edges of the facing surfaces of the first and second tree engaging structures when the fastener draws the facing surface against the tree. The ramped surfaces of the first and second tree engaging structures facilitate the sliding of the first and second tree engaging structures down the tree.

(49) An example climbing stick includes a frame, a step, a first tree engaging structure, a second tree engaging structure, and a fastener. The step is coupled to the frame and is configured to support a user standing thereon. The first tree engaging structure can be fixed at a first end the frame and includes a facing surface. The facing surface of the first tree engaging structure has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge. The second tree engaging structure can be fixed at a second end the frame and includes a facing surface. The facing surface of the second tree engaging structure has a top edge, a bottom edge, and a ramped surface extending between the top edge and the bottom edge. The fastener is configured to attach to the frame and to draw the facing surfaces of the first and second tree engaging structures against the tree. The top edges of the facing surfaces of the first and second tree engaging structures are disposed closer to a center of the tree than the bottom edges of the facing surfaces of the first and second tree engaging structures when the fastener draws the facing surface against the tree. The ramped surfaces of the facing surfaces of the first and second tree engaging structures facilitate the sliding of the first tree engaging structure down the tree.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention is described with reference to the following drawings, wherein like reference numbers denote substantially similar elements;
- (2) FIG. 1 is a perspective view of a portable treestand and climbing stick system mounted on a tree;
- (3) FIG. 2 is a perspective view of the system of FIG. 1 in a stowed position;
- (4) FIG. 3 is an exploded perspective view of the treestand of FIG. 1;
- (5) FIG. 4A is a top perspective view of the platform of the treestand of FIG. 1;
- (6) FIG. 4B is a bottom perspective view of the platform of the treestand of FIG. 1;
- (7) FIG. 5 is an exploded perspective view of a bow engaging device of the treestand of FIG. 1;
- (8) FIG. 6 is a close-up perspective view of the bow engaging device of FIG. 5;
- (9) FIG. 7 is a cross-sectional view of a portion of the platform of the treestand of FIG. 1 with a bow resting therein;
- (10) FIG. 8A is a plan view of a front wall of the frame of the treestand of FIG. 1;
- (11) FIG. 8B is a plan view of a right side wall of the frame of the treestand of FIG. 1;
- (12) FIG. 8C is a plan view of a rear wall of the frame of the treestand of FIG. 1;
- (13) FIG. 8D is a plan view of a left side wall of the frame of the treestand of FIG. 1;
- (14) FIG. 9 is an exploded perspective view of a seat assembly of the treestand of FIG. 1;
- (15) FIG. 10 is a top view of a tree engaging structure of the treestand of FIG. 1;
- (16) FIG. 11 is a perspective view of a hinge bracket fixed to the platform of the treestand of FIG. 1 according to an alternate embodiment;
- (17) FIG. 12A is a side view of the treestand of FIG. 1 in a deployed position;

- (18) FIG. 12B is a side view of the treestand of FIG. 1 transitioning from the deployed position to a collapsed position;
- (19) FIG. 12C is a side view of the treestand of FIG. 1 in the collapsed position;
- (20) FIG. 13 illustrates the forming of the platform of FIG. 4A from a solid plate;
- (21) FIG. 14 illustrates the forming of the seat of FIG. 9 from a solid plate;
- (22) FIG. 15 illustrates the forming of two of the platforms of FIG. 4A and two of the seats of FIG. 9 from a solid plate;
- (23) FIG. 16 is a perspective view of an alternate platform;
- (24) FIG. 17 is a perspective view of a climbing treestand;
- (25) FIG. 18 is a top view of an alternate tree engaging structure;
- (26) FIG. 19 is a flowchart summarizing an example method of manufacturing a treestand;
- (27) FIG. 20 is a flowchart summarizing an example method of performing the third step (provide monolithic platform) of the method of FIG. 19;
- (28) FIG. 21 is an exploded perspective view of a climbing stick of FIG. 1;
- (29) FIG. 22 is a perspective view of a step structure of FIG. 21;
- (30) FIG. 23 is a plan view of the step structure of FIG. 22;
- (31) FIG. 24 is a plan view of a tree engaging feature of FIG. 21;
- (32) FIG. 25A is a plan view of a front wall of a frame of the climbing stick of FIG. 21;
- (33) FIG. 25B is a plan view of a right side wall of the frame of the climbing stick of FIG. 21;
- (34) FIG. 25C is a plan view of a rear wall of the frame of the climbing stick of FIG. 21;
- (35) FIG. 25D is a plan view of a left side wall of the frame of the climbing stick of FIG. 21;
- (36) FIG. 26 is a top plan view of the climbing sticks of FIG. 21 being stacked onto one another;
- (37) FIG. 27 is a side plan view of the climbing sticks of FIG. 21 being stacked onto one another;
- (38) FIG. 28 is a perspective view of two of the climbing sticks of FIG. 21 being stacked onto one another;
- (39) FIG. 29 is a perspective view of an alternate climbing stick;
- (40) FIG. 30 is a perspective view of a step structure of the climbing stick of FIG. 29;
- (41) FIG. 31 is a perspective view of another alternate climbing stick;
- (42) FIG. 32 is a perspective view of a step structure of the climbing stick of FIG. 31;
- (43) FIG. 33 is a flowchart summarizing a method of manufacturing a climbing stick;
- (44) FIG. 34 is a rear perspective view of an alternate climbing stick;
- (45) FIG. 35 is a perspective view of a bottom surface of a tree engaging structure of the alternate climbing stick of FIG. 34;
- (46) FIG. 36 is a side plan view of the tree engaging structure of FIG. 35;
- (47) FIG. 37A is a side plan view of the alternate climbing stick of FIG. 34 mounted to a tree and having a small force imparted thereon;
- (48) FIG. 37B is a side plan view of the alternate climbing stick of FIG. 34 mounted to a tree and having a larger force imparted thereon;
- (49) FIG. 38 is a rear perspective view of a bottom surface of an alternate tree engaging structure;
- (50) FIG. 39 is a bottom plan view of the alternate tree engaging structure of FIG. 38;
- (51) FIG. 40 is a side plan view of an alternate treestand mounted to a tree;
- (52) FIG. 41A is a side plan view of an alternate tree engaging structure;
- (53) FIG. 41B is a side plan view of another alternate tree engaging structure;
- (54) FIG. 41C is a side plan view of yet another alternate tree engaging structure;
- (55) FIG. 41D is a side plan view of yet another alternate tree engaging structure; and
- (56) FIG. 41E is a side plan view of yet another alternate tree engaging structure.

DETAILED DESCRIPTION

(57) The present invention overcomes the problems associated with the prior art, by providing a more portable treestand and climbing stick system. In the following description, numerous specific details are set forth (e.g., material types, fastener types, etc.) in order to provide a thorough

understanding of the invention. Those skilled in the art will recognize, however, that the invention may be practiced apart from these specific details. In other instances, details of well-known manufacturing practices (e.g., extruding, welding, etc.) and components have been omitted, so as not to unnecessarily obscure the present invention.

(58) FIG. 1 shows a perspective view of a portable treestand and climbing stick system **100** mounted on a tree **102**. System **100** includes a treestand **104**, a plurality of climbing sticks **106**, and a plurality of cam straps **108**. Treestand **104** is configured to be mounted up in a tree to provide a user with a secure structure to stand and sit on. Treestand **104** is shown deployed and fastened to tree **102** with two of straps **108**. Climbing sticks **106** are configured to be removably mounted to tree **102** so that a user can climb into, and out of, treestand **104**. Each of climbing sticks **106** is secured to tree **102** with a respective one of straps **108**. Straps **108** are, for example, lightweight nylon cam straps having an adjustable length to accommodate for trees having varying trunk diameters. Each end of strap **108** includes a closed loop adapted to easily engage and disengage small posts structures of treestand **104** and/or climbing sticks **106**.

(59) FIG. 2 shows a perspective view of system **100** in a stowed position, with treestand **104** in a collapsed position and four of sticks **106** stacked and secured to treestand **104** with a set of bungee cords **200**. Alternatively, bungee cords **200** can be omitted and sticks **106** can be secured to treestand **104** using straps **108**. Although not shown, backpack straps can be fastened to the bottom of treestand **104**, so that system **100** can be carried on the back of a user.

(60) FIG. 3 shows an exploded perspective view of treestand **104** according to one embodiment of the present invention. Treestand **104** includes a platform **300**, a hinge bracket **302**, a frame **304**, a cable assembly **306**, a seat assembly **308**, a tree engaging structure **310**, and a set of posts **312**.

(61) Platform **300** provides a relatively planar surface upon which a user can stand when treestand **104** is deployed and secured to a tree.

(62) Bracket **302** facilitates the pivoting of platform **300** with respect to frame **304**. Bracket **302** is seated and fixed in a recess **314** of platform **300**. In the example embodiment, bracket **302** is fixed directly to platform **300** with a set of screws **317**.

(63) Frame **304** provides structural support to treestand **104** and is hingably connected to bracket **302** with a bolt **316** and a nut **318**. A set of oil-infused nylon washers **320** are disposed between bracket **302** and frame **304**, to eliminate metal-on-metal contact therebetween. By eliminating metal-on-metal contact, noise is minimized when platform **300** is pivoted with respect to frame **304**.

(64) Cable assembly **306** provides tensile support to platform **300** when treestand **104** is deployed. Cable assembly **306** includes a set of steel cables **322**, each having a first end **324** and a second end **326**. Each of first ends **324** is fastened to platform **300** with a respective bolt **328** and a nut **330**. Second ends **326** are adjustable secured to frame **304** with an adjustable shaft assembly **332**, which facilitates the leveling of platform **300** with respect to frame **304**.

(65) Seat assembly **308** includes a seat **334** and a leveling system **336**. Seat **334** is hingably connected to frame **304** with a bolt **338** and a nut **340**. A second set of oil-infused nylon washers **342** are disposed between seat **334** and frame **304**, to eliminate metal-on-metal contact therebetween. Leveling system **336** facilitates the leveling of seat **334** with respect to frame **304**. A polymer pad **344** is attached to frame **304** and is adapted to abut leveling system **336**.

(66) In the example embodiment, treestand **104** further includes a bow engaging device **346** and polymer pad **348** for securing a bow (i.e., an archery bow) to platform **300**. Bow engaging device **346** is adjustably secured to platform **300** with two set screws **350**. Polymer pad **348** provides a soft surface that abuts the limb of the bow.

(67) Tree engaging structure **310** is mounted to the rear side of frame **304**, to facilitate the mounting of treestand **104** to a tree. Tree engaging structure **310** is configured to bite into the tree, to prevent treestand **104** from slipping against the tree. Furthermore, tree engaging structure **310** is adapted engage trees of varying sizes and also helps to secure frame **304** in a vertical position when

treestand **104** is mounted to a section of a tree that is not exactly vertical. Tree engaging structure **310** is mounted directly to frame **304** with a bolt **352** and a complementary nut **354**.

(68) Posts **312** provide a structure onto which end loops of straps **108** are positioned when securing treestand **104** to a tree. Each of posts **312** is attached to frame **304** with a respective bolt **356** and complementary nut **358**. In this example, each of posts **312** is machined from black DELRIN™ polymer.

(69) FIG. 4A is a perspective view of platform **300**, which is a monolithic structure. In the example embodiment, platform **300** is computer numeric control (CNC) machined from a single solid plate of 6061 aluminum alloy. Before platform **300** is cut from the solid plate, the plate is subjected to pre strengthening and/or toughening processes including, but not limited to, metal working processes (e.g., cold working/rolling, hot working/rolling, extrusion, forging, etc.) and/or temperature treatment processes (e.g., tempering, annealing, quenching, case hardening, precipitation strengthening, normalizing, etc.). Because the plate is subjected to one or more of such processes, platform **300** has several advantages over prior art cast platforms. For example, platform **300** is lighter than cast platforms because the plate from which platform **300** is cut is pre-strengthened/toughened, thus requiring less material to support the same load. Another reason for this is that platform **300** is more ductile than a cast platform and, therefore, not as susceptible to sudden, catastrophic fracture. Because treestands support people at relatively high elevations, sudden fracture of a brittle platform is highly undesirable. To avoid such catastrophic failures, cast platforms have to be over-engineered. Platform **300**, however, is more ductile and tough and, therefore, much more likely to deform (e.g., bend) before failing (e.g., shattering). This allows platform **300** to have a much more relaxed safety factor. Indeed, platform **300** is lighter, stronger, and safer than larger platforms that are cast from similar metals and designed to support the same load.

(70) Platform **300** includes a plurality of peripheral sidewalls **400**, a plurality of structural supports **402**, and a tree engaging feature **404**.

(71) In the example embodiment, the height of peripheral sidewalls **400** is the same as the thickness of the plate from which platform **300** is cut from. The left and right portions of sidewalls **400** each define a respective aperture **406** passing completely therethrough. Each of apertures **406** is adapted to receive a respective one of bolts **328** to facilitate the fastening of cables **322** to platform **300**. Because apertures pass through sidewalls **400**, no upward projections are required to connect cables **322** to platform **300**, thereby reducing the required thickness of the metal plate from which platform **300** is cut.

(72) Structural supports **402** define a plurality openings **408** therebetween. Openings **408** pass completely through platform **300** such that structural supports **402** are arranged in a web configuration. The web configuration is intentionally irregular shaped so that platform **300** blends in with irregular shaped surrounding structures such as branches, limbs, etc. This provides a camouflage effect and prevents game from noticing the presence of treestand **104**. As shown, the height of supports **402** is slightly less than the thickness of the plate from which platform **300** is formed. Each of supports **402** includes a plurality of channels **410**, machined therein, to provide additional traction and prevent the foot of a user from slipping on platform **300**. Structural supports **402** also define an integral bow receiving feature **412**, which receives the bottom limb and cam of a compound bow. The structural supports **402** defining bow receiving feature **412** include a plurality of threaded apertures **414** passing therethrough. Apertures **414** are threaded to receive set screws **350** of bow engaging device **346**. The number of apertures **414** allow bow engaging device **346** to be adjustably positioned at any two of apertures **414** to compensate for different sized bows. Furthermore, apertures **414** are positioned on both sides of bow receiving feature **412** thus allowing bow engaging device **346** to be set up for either left or right handed bows/hunters.

(73) Tree engaging feature **404** is an integral part of platform **300** and facilitates the mounting of treestand **104** to a tree. Feature **404** is angled to accommodate for varying width trees and defines a

plurality of teeth **416** configured to grip/bite into the side of a tree when treestand **104** is fastened thereto.

(74) Recess **314** is formed in an intermediate portion **417** of platform **300**, disposed between tree engaging feature **404** and the web of support structures **402**. Because platform **300** is machined from a single piece of pre-strengthened/toughened metal, the width of intermediate portion can be significantly narrowed, resulting in substantial weight reduction. Indeed, in this example embodiment, intermediate portion **417** is only slightly (e.g., 10%) wider than recess **314** and significantly less than twice the cross-sectional dimension of tubular frame **304**.

(75) In the example embodiment, platform **300** further includes a plurality of recesses **418** that do not pass completely through platform **300**. Retaining a thin portion of the bottom surface of platform **300** within the recesses **418** provides significant weight reduction while still maintaining a high structural strength immediately adjacent intermediate portion **417**.

(76) FIG. **4B** shows a rear perspective view of platform **300**. As shown, two of the four recesses **418** (shown in FIG. **4A**) include hoof-shaped openings **420** passing completely therethrough. The hoof-shaped openings resemble deer tracks and are considered to be aesthetically pleasing to hunters.

(77) FIG. **5** shows an exploded perspective view of bow engaging device **346**, which includes a body **500**, a pair of set screws **502**, a polymer tube **504**, and a hex screw **506**. Body **500** is a generally c-shaped metal structure defining a channel **508**, a set of threaded apertures **510**, and another threaded aperture **512**. Channel **508** is configured to receive a structural support **402** of platform **300**. Threaded apertures **510** receive set screws **502**. During the mounting of body **500** to one of structural supports **402**, apertures **510** are coaxially aligned with a selected pair of apertures **414**. Once aligned, each of screws **502** is threaded into both a respective aperture **510** and a respective aperture **414**, thereby fastening body **500** to platform **300**. Accordingly, apertures **414** and **510** have the same thread specifications. Threaded aperture **512** receives screw **506**. Before screw **506** is threaded into aperture **512**, tube **504** is disposed around the shaft of screw **506**. Tube **504** provides a relatively soft surface that effectively engages a bow without damaging it and/or causing noise.

(78) FIG. **6** shows a close-up perspective view of bow engaging device **346** and polymer pad **348** attached to platform **300**.

(79) FIG. **7** is a cross-sectional view of platform **300** taken along line A-A of FIG. **6** wherein a bow **700** is shown seated in bow receiving feature **412**. As shown, pad **348** abuts the bottom limb **702** of bow **700** while tube **504** abuts a cam **704** and a string **706** of bow **700**. The center of mass of bow **700** is in front of pad **348** and, therefore, bow **700** has a tendency to lean forward (to the left in FIG. **7**). Bow **700** is prevented from tipping forward by the downward force exerted on cam **704** by tube **504**.

(80) FIGS. **8A-8D** show plan views of the front wall **800**, right side wall **802**, rear wall **804**, and left side wall **806**, respectively, of frame **304**. In the example embodiment, frame **304** is formed from a section of extruded aluminum square tube.

(81) Front wall **800** of frame **304**, shown in FIG. **8A**, includes a first set of apertures **808**, a second aperture **810**, and a plurality of weight reduction apertures **812**, all passing completely therethrough. Apertures **808** provide passages through which bolts **356** are disposed to mount posts **312** to frame **304**. Aperture **810** provides a passage through which bolt **352** is disposed to mount tree engaging structure **310** to frame **304**. Apertures **812** reduce the overall weight of frame **304** while also making it more difficult for animals to see frame **304**, when treestand **104** is mounted in a tree. In this example, each of apertures **812** includes a 45 degree chamfer, which further reduces the weight of frame **304**.

(82) The right side wall **802** of frame **304**, shown in FIG. **8B**, includes a first aperture **814**, a second aperture **816**, a slotted cutout **818**, and additional weight reduction apertures **812**, all passing completely therethrough. Aperture **814** provides a passage through which bolt **316** is disposed to

hingably mount frame **304** to bracket **302**. Aperture **816** provides a passage through which bolt **338** is disposed to hingably mount seat **334** to frame **304**. Cutout **818** provides multiple downward angled channels that allow a user to optionally change the vertical position of adjustable shaft assembly **332**, which connects cables **322** to frame **304** (FIG. 3). The vertical position of adjustable shaft assembly **332**, therefore, dictates the angle of platform **300** with respect to frame **304**. For example, if treestand **104** is mounted in a rear leaning tree, platform **300** can be leveled by moving adjustable shaft assembly **332** to a lower one of the downward angled channels, thereby increasing the angle between platform **300** and frame **304**.

(83) Rear wall **804** is similar to front wall **800**. The only difference being that rear wall **804** includes apertures **820**, which are coaxially aligned with apertures **808**, respectively, but are much larger. The larger diameter of apertures **820** allow a tool (e.g. socket) to pass through rear wall **804** during the fastening of nuts **358** onto bolts **356** (FIG. 3). As shown, aperture **810** passes through both front wall **800** and rear wall **804**.

(84) Left side wall **806** is a mirror reflection of right side wall **802**. As shown, aperture **814**, aperture **816**, slotted cutout **818**, and weight reduction apertures **812** pass completely through both right side wall **802** and left sidewall **806**.

(85) One advantageous aspect of this example embodiment is that weight reduction apertures **812** are offset between adjacent walls. For example, the vertical center points of apertures **812** formed through front wall **800** and rear wall **804** are located halfway between the vertical center points of apertures **812** formed through right side wall **802** and left side wall **806**. This offset allows the maximum number of weight reduction apertures **812** to be formed in frame **304**, while sacrificing only minimal structural strength.

(86) FIG. 9 shows an exploded perspective view of seat assembly **308**. In the example embodiment, seat **334** is a monolithic structure formed from a single solid plate of 6061 aluminum alloy. Before seat **334** is cut from the solid plate, the plate is subjected to a strengthening process. This allows seat to be stronger and lighter than prior art seats. As a result, seat **334** can be made thinner and lighter than prior art seats.

(87) Seat **334** includes a plurality of structural supports **900**, a set of threaded apertures **902**, and a set of smooth apertures **904**. Structural supports **900** are similar to structural supports **402** of platform **300** in that they define a plurality openings **906** therebetween. Openings **906** pass completely through seat **334**, so that structural supports **900** are arranged in a web configuration. This allows seat **334** to be as light as possible while still having sufficient structural strength to support a user seated thereon, when treestand **104** is mounted in a tree. Threaded apertures **902** are adapted to receive a set of screws **908** which fasten leveling system **336** to the rear surface of seat **334**. Smooth apertures **904** are adapted to receive the smooth exterior shaft of bolt **338** so as to hingably couple seat **334** to frame **304**. Accordingly, apertures **904** coaxially align with aperture **816** of frame **304**.

(88) Leveling system **336** provides a means for changing the angle between seat **334** and frame **304**, to accommodate for non-vertical trees and/or user preference. Leveling system **336** includes a bracket **910** and a screw **912**. Bracket **910** includes a set of apertures **914** and a threaded aperture **916**. Apertures **914** are adapted to receive screws **908** and facilitate the mounting of bracket **910** to seat **334**. Aperture **916** is adapted to receive screw **912** such that threading screw **912** into aperture **916** causes the angle between seat **334** and frame **304** to increase. Oppositely, loosening screw **912** causes the angle between seat **334** and frame **304** to decrease. The tip of screw **912** urges against polymer pad **344** when a user sits in seat **334**.

(89) FIG. 10 shows a top view of tree engaging structure **310** removed from frame **304**. Tree engaging structure **310** is an aluminum structure formed, for example, by first extruding the antler-shaped profile along an axis into a beam format and then cutting the beam perpendicular to the axis. Accordingly, a single beam yields multiple tree engaging structures **310**.

(90) Tree engaging structure **310** includes a channel **1000**, an aperture **1002**, a set of outside teeth

1004, a set of middle teeth **1006**, and a set of inside teeth **1008**. Channel **1000** is adapted to seat frame **304**, to prevent tree engaging structure **310** from rotating about bolt **352** (FIG. 3). Channel **1000** is defined by a first surface **1010**, a second surface **1012**, and a third surface **1014**. When tree engaging structure **310** is mounted to frame **304**, first surface **1010** engages rear wall **804** of frame **304**, second surface **1012** engages right sidewall **802** of frame **304**, and third surface **1014** engages left sidewall **806** of frame **304**. Aperture **1002** is adapted to receive bolt **352** and coaxially aligns with aperture **810** of frame **304**. Teeth **1004**, **1006**, and **1008** are configured to bite into the bark of a tree. In the example embodiment, the points of teeth **1004** and **1006** are coplanar while the points of teeth **1008** are slightly set back closer to channel **1000**. During the mounting of treestand **104** to a tree, only two of the six teeth **1004**, **1006**, and **1008** need to bite into the tree. The number and configuration of teeth **1004**, **1006**, and **1008** is such that tree engaging structure **310** can engage various size and shaped trees. This is beneficial in that tree engaging structure **310** can engage angled trees while still securing frame **304** in a vertical upright position. Optionally, teeth **1004** can be made sharper than teeth **1006**, so that teeth **1004** can penetrate deeper into a tree than teeth **1006**, to accommodate the round contour of a non-vertical tree, while maintaining treestand **104** in a substantially level position.

(91) FIG. **11** shows a perspective view of hinge bracket **302** fixed to platform **300** according to another embodiment of the present invention. In this particular embodiment, bracket **302** is fixed to platform **300** via a weld joint **1100** rather than by screws **317**.

(92) FIGS. **12A-12C** show side views of treestand **104** as it is transitioned from a deployed position to a collapsed position. More specifically, FIG. **12A** shows a side view of treestand **104** in a deployed position, FIG. **12B** shows a side view of treestand **104** transitioning from a deployed position to a collapsed position, and FIG. **12C** shows a side view of treestand **104** in a collapsed position.

(93) FIG. **13** illustrates forming of platform **300** from a solid plate **1300** of 6160 aluminum alloy. In this example, plate **1300** is strengthened/toughened before it is CNC machined into the shape of platform **300**.

(94) FIG. **14** illustrates forming of seat **334** from a solid plate **1400** of 6160 aluminum alloy. In this example, plate **1400** is CNC machined into the shape of seat **334**.

(95) FIG. **15** illustrates the forming of multiple platforms **300** and seats **334** from a single solid 6061 aluminum plate. By forming multiple platforms **300** and seats **334** from a single plate **1500**, there is less scrap aluminum plate as compared to forming each part from a separate plate. This is because the platforms **300** and seats **334** can be collectively arranged to more efficiently occupy plate real-estate.

(96) In the example embodiments shown in FIGS. **13-15**, platform **300** and seat **334** are machined (milling, drilling, etc.) from plates using a CNC machine. However, other cutting means can be implemented in place of machining. For example, platform **1300** and seat **334** can be cut from a plate using other processes including, but not limited to, water jet cutting, plasma cutting, flame cutting, laser cutting, stamping/punching, drilling, and/or any other manufacturing process for removing material. Such alternate cutting processes can optionally be automated, using CNC or other systems, or manual.

(97) FIG. **16** is a perspective view of a platform **1600** according to an alternate embodiment of the present invention. Platform **1600** is substantially similar to platform **300** of treestand **100**, except that platform **1600** includes an antler-shaped tree engaging feature **1602**. Indeed, platform **1600** can be used in place of platform **300** in treestand **104**. Features of platform **1600** that are identical to those of platform **300** are denoted with like reference numbers and their descriptions are withheld from the description of FIG. **16** to avoid redundancy.

(98) Antler-shaped tree engaging feature **1602** (as well as tree engaging structure **310** shown in FIG. **10**) includes both functional and ornamental features. In particular, the shape of tree engaging feature **1602** resembles the antlers (rack) of a deer and is, therefore, thought to be aesthetical

pleasing to hunters. In addition, the tines of the antler design are effective to penetrate the bark of a tree and prevent unwanted movement with respect to the tree.

(99) FIG. 17 is a perspective view of a climbing treestand 1700 according to another alternate embodiment of the present invention. Treestand 1700 includes a platform assembly 1702 and a seat assembly 1704, both of which are configured to be fastened to a tree. Platform assembly 1702 includes a platform 1706 and an adjustable strap assembly 1708 coupled thereto. Platform 1706 includes an integral tree engaging feature 1710. Seat assembly 1704 includes a seat 1712 and an adjustable strap assembly 1714 coupled thereto. Seat 1712 also includes an integral tree engaging feature 1716. Platform 1706 and seat 1712 are each machined from a solid plate (either the same plate or different plates) of aluminum alloy or other pre-strengthened metal.

(100) FIG. 18 shows a top view of an alternate tree engaging structure 1800 for use with treestand 104. Tree engaging structure 1800 includes a channel 1802, an aperture 1804, a set of outside teeth 1806, a set of middle teeth 1808, and a set of inside teeth 1810. Channel 1802 is adapted to receive/seat frame 304 to prevent tree engaging structure 1800 from rotating about bolt 352.

Channel 1802 is defined by a first surface 1812, a second surface 1814, and a third surface 1816. When tree engaging structure 1800 is mounted to frame 304, first surface 1812 engages rear wall 804 of frame 304, second surface 1814 engages right sidewall 802 of frame 304, and third surface 1816 engages left sidewall 806 of frame 304. Aperture 1804 is adapted to receive bolt 352 and coaxially aligns with aperture 810 of frame 304. Teeth 1806, 1808, and 1810 are configured to bite into the bark of a tree. In the example embodiment, the points of teeth 1806 and 1808 are coplanar while the points of teeth 1810 are slightly set back closer to channel 1802. It is important to note that outside teeth 1806 are substantially sharper than middle teeth 1808. Therefore, outside teeth 1806 can bite deeper into the tree than do middle teeth 1808. Because outside teeth 1806 bite deeper into the tree, tree engaging structure 1800 can engage substantially angled trees while still securing frame 304 in a vertical upright position.

(101) FIG. 19 is a flowchart summarizing an example method 1900 of manufacturing a treestand. In a first step 1902, a seat is provided. Then, in a second step 1904, a tree engaging structure is provided. Next, in a third step 1906, a monolithic platform is provided. Finally, in a fourth step 1908, the seat, tree engaging structure, and the monolithic platform are combined to form a treestand.

(102) FIG. 20 is a flowchart summarizing an example method 2000 of performing third step 1906 (provide monolithic platform) of method 1900. In a first step 2002, a solid piece of material (e.g., pre-strengthened or pre-toughened metal) is provided. Then, in a second step 2004, portions of the solid piece of material are removed to form openings between a top and bottom surface of the platform defining structural supports therebetween.

(103) FIG. 21 shows an exploded perspective view of one of climbing sticks 106. Climbing stick 106 includes a set of step structures 2100, a set of feet 2102, a post 2104, and a set of end caps 2106 coupled to a frame 2108. Step structures 2100 provide a surface for a user to grab and stand on when ascending and descending a tree via climbing stick 106. Feet 2102 are essentially tree engaging structures which are configured to bite into a tree to maintain a fixed relationship between climbing stick 106 and the tree. Post 2104 provides a structure for a strap 108 (shown in FIG. 1) to engage when securing climbing stick 106. End caps 2106 are disposed in opposite ends of frame 2108 for aesthetic purposes and to cover the sharp edges of frame 2108. Frame 2108 is a tubular structure with a square cross-section, and provides the main structure to which all other components of climbing stick 106 are mounted.

(104) Each of step structures 2100 is fixably mounted to frame 2108 by a top bolt 2110, a top nut 2112, a bottom bolt 2114, and a bottom nut 2116. Bolts 2110 are coaxially disposed through a top aperture 2118 of step structure 2100, an aperture 2120 of frame 2108, and finally through nut 2112. Bolts 2114 also facilitate the mounting of step structures 2100 and feet 2102 to frame 2108. Bolts 2114 are coaxially disposed through a bottom aperture 2124 of step structures 2100, an aperture

2126 of frame **2108**, aperture **2122** of feet **2102** and finally through nut **2116**.

(105) Post **2104** provides a structure onto which each loop of a strap **108** engages during the mounting of climbing stick **106** to a tree. Post **2104** is mounted to frame **2108** with a bolt **2128** and nut **2130**. More specifically, bolt **2128** is disposed through an aperture **2132** of post **2104**, an aperture **2134** of frame **2108**, and finally through nut **2130**.

(106) FIG. **22** shows a rear perspective view of one of step structures **2100**. In addition to apertures **2118** and **2124**, step structure **2100** further includes a channel **2200**, a set of protrusions **2202**, a plurality of grooves **2204**, two slots **2206**, and a plurality of weight reduction apertures **2208**. Channel **2200** is adapted to receive and seat the outer surface of frame **2108** so as to distribute the forces exerted by step structure **2100** on bolts **2110**, **2114**, and frame **2108**, when a user stands on step structure **2100**. Channel **2200** also prevents relative movement (e.g., rotational and/or translational movement) between step structure **2100** and frame **2108**. Protrusions **2202** also engage opposite sides of frame **2108** to resist the torque exerted on step structure **2100** when a user stands on step structure **2100**. Protrusions **2202** also function as restraining features configured to prevent the tree engaging structures of another climbing stick from sliding along frame **2108** when climbing sticks **106** are stacked onto one another. Slots **2206** allow feet **2102** of other climbing sticks **106** to engage frame **2108**, when stacking climbing sticks **106** onto one another. Grooves **2204** provide traction to the boot of a user standing on step structure **2100**, and apertures **2208** reduce the weight of step structures **2100** without sacrificing significant structural strength.

(107) FIG. **23** shows a rear view of one of step structures **2100**. As shown, each side **2300** of step structure **2100** is slightly pitched upward at an angle of twelve degrees with respect to the horizontal. This provides additional grip between the boot of a user and step structure **2100**. In the example embodiment, each of step structures **2100** is formed first extruding an aluminum beam having a cross-sectional profile as shown in FIG. **23**. Then, the beam is cut perpendicularly with respect to the extrusion axis of the beam. Finally, channel **2200** is machined into each step structure **2100**.

(108) FIG. **24** shows a top plan view of one of feet **2102**. Each of feet **2102** further includes a plurality of antler-shaped teeth **2400**, a channel **2402**, apertures **2404**, and a channel **2406**. Teeth **2400** are configured to bite into the outer surface of a tree to facilitate the mounting of climbing stick **106** to the tree. Channel **2402** is adapted to receive the outer surface of frame **2108** to prevent relative rotation of feet **2102** with respect to frame **2108**. That is, each side **2408** of channel **2402** engages a respective opposite side of frame **2108**. Apertures **2404** reduce the overall weight of climbing sticks **106**. Channel **2406** is a frame engaging feature that is adapted to receive the frame **2108** of another climbing stick **106**, when climbing sticks **106** are stacked on one another. The stacking of climbing sticks **106** will be shown in greater detail in FIG. **26**. FIGS. **25A-25D** show plan views of the front wall **2500**, right side wall **2502**, rear wall **2504**, and left side wall **2506**, respectively, of frame **2108**. In the example embodiment, frame **2108** is formed from a section of extruded aluminum square tube.

(109) Front wall **2500** of frame **2108** shown in FIG. **25A** includes apertures **2120**, apertures **2126**, aperture **2134** and a plurality of weight reduction apertures **2508**, all passing completely therethrough. Apertures **2120** provide passages through which bolts **2110** are disposed to mount step structures **2100** to frame **2108**. Apertures **2126** provides a passage through which bolts **2114** are disposed to mount step structures **2100** to frame **2108** and to also mount feet **2102** to frame **2108**. Aperture **2134** receives bolt **2128** to mount post **2104** to frame **2108**. Apertures **2508** reduce the overall weight of frame **2108** while also making it more difficult for animals to see frame **2108** when climbing sticks **106** are mounted to a tree. In this example, each of apertures **2508** includes a 45 degree chamfer, which further reduces the weight of frame **2108**.

(110) The right side wall **2502** of frame **2108** shown in FIG. **25B** also includes a plurality of chamfered weight reduction apertures **2508**.

(111) Rear wall **2504** is similar to front wall **2500**, except that rear wall **2504** includes apertures

2510 and **2512**. Apertures **2510** and **2512** are coaxially aligned with apertures **2120** and **2134**, respectively, but are larger. The larger diameter of apertures **2510** and **2512** allow a tool (e.g. socket) to pass through rear wall **2504** during the fastening of nuts **2116** and **2130** onto bolts **2110** and **2128**, respectively. As shown, apertures **2126** pass through both front wall **2500** and rear wall **2504**.

(112) Left side wall **2506** is identical to right side wall **2500** and is, therefore, not explained in detail to avoid redundancy.

(113) One important aspect of example climbing stick **106** is that weight reduction apertures **2508** are offset with respect to apertures **2508** on adjacent walls. For example, the vertical center points of apertures **2508** formed through front wall **2500** and rear wall **2504** are located halfway between the vertical center points of apertures **2508** formed through right side wall **2502** and left side wall **2506**. This offset allows an increased number of weight reduction apertures **2508** to be formed in frame **2108**, while sacrificing only minimal structural strength.

(114) FIG. **26** shows a top plan view of climbing sticks **106** being stacked onto one another. As shown, each of channels **2406** is adapted to engage the frame **2108** of an underlying one of climbing sticks **106**.

(115) FIG. **27** shows a side view of climbing sticks **106** being stacked onto one another.

(116) Climbing sticks **106** are configured to stack on to one another in a substantially horizontally aligned position, when frames **2108** are oriented horizontally. In other words, climbing sticks **106** are configured to engage one another in an aligned position such that corresponding features of climbing sticks **106** are substantially aligned along an axis **2700**. Axis **2700** is perpendicular to an axis **2702** of frame **2108**, as well as to the axes of the other stacked frames **2108**. In this example, climbing sticks **106** stack directly on top of one another such that bolts **2114** of each respective one of climbing sticks **106** are substantially coaxially aligned with one another within a predetermined tolerance. Horizontal alignment provides an important advantage by reducing the size (e.g., the width in FIG. **27**) of the stack of climbing sticks **106**, making the stack of climbing sticks **106** easier to carry through thick, tangled brush.

(117) FIG. **28** shows a perspective view of one of climbing sticks **106** being stacked onto another one of climbing sticks **106**. When stacked, channel **2406** of the top climbing stick **106** engages the portion of frame **2108** that is exposed by slots **2116** of step **2000** of the underlying climbing stick **106**. Protrusions **2202** of the underlying climbing stick **106** engage the tree engaging structure **2102** of the climbing stick **106** stacked thereon, thereby preventing the top climbing stick **106** from sliding along frame **2108** of the underlying climbing stick **106**.

(118) FIG. **29** shows a perspective view of an alternate climbing stick **2900**. Climbing stick **2900** includes a frame **2902**, a post **2904**, and a pair of step structures **2906**. Post **2904** provides a structure that straps **108** can engage, when mounting climbing stick **2900** in a tree. Post **2904** is mounted to frame **2902** with a bolt **2908**. Each of step structures **2906** is mounted at an opposite end of frame **2902** with a respective bolt **2910**.

(119) FIG. **30** shows a perspective view of one of step structures **2906** removed from frame **2902**. Step structure **2906** includes an integral tree engaging feature **3000**, a frame receiving aperture **3002**, a bolt receiving aperture **3004**, and a step feature **3006**. Tree engaging feature **3000** is configured to bite into a tree and to facilitate the stacking of climbing sticks **2900** onto one another. Frame receiving aperture **3002** is configured to receive an end of frame **2902**. Aperture **3004** passes completely through step structure **2906** such that bolt **2910** can be disposed simultaneously through both step structure **2906** and frame **2902**. Step structure **2906** is a monolithic structure including tree engaging feature **3000** and step feature **3006** as integral parts thereof. The monolithic nature of step structure **2906** provides important advantages including, but not limited to, simpler assembly, fewer parts, improved stability, and increased strength.

(120) FIG. **31** shows a perspective view of another alternate climbing stick **3100**. Climbing stick **3100** includes a frame **3102**, a post **3104**, and a pair of step structures **3106**. Post **3104** provides a

structure which straps **108** can engage, when mounting climbing stick **3100** to a tree. Post **3104** is mounted to frame **3102** with a bolt **3108**. Each of step structures **3106** is mounted at an opposite end of frame **3102** with a respective bolt **3110**.

(121) FIG. **32** shows a perspective view of one of step structures **3106** removed from frame **3102**. Step structure **3106** includes an integral tree engaging feature **3200**, a frame receiving protrusion **3202**, a bolt receiving aperture **3204**, and a step feature **3206**. Tree engaging feature **3200** is configured to bite into a tree and to facilitate the stacking of climbing sticks **3100** onto one another. Frame receiving protrusion **3202** is adapted to slide into and fit closely within an end of frame **3102**. Aperture **3204** passes completely through protrusion **3202** such that bolt **3110** can be disposed simultaneously through both step structure **3100** and frame **3102**. Step structure **3106** is a monolithic structure including tree engaging feature **3200** and step feature **3206** as integral parts thereof. The monolithic nature of step structure **3106** provides important advantages similar to step structure **2906** of FIG. **30**.

(122) FIG. **33** is a flowchart summarizing a method **3300** of manufacturing a climbing stick. In a first step **3302**, a frame tube is provided. Then, in a second step **3304**, weight reduction apertures are formed in the frame tube. Next, in a third step **3306**, a step structure is provided. Then, in a fourth step **3308**, the step structure is coupled to the frame tube. Next, in a fifth step **3310**, a tree engaging structure is provided. Finally, in a sixth step **3312**, the tree engaging structure is coupled to the frame tube. Optionally, the step structure and tree engaging feature are integral parts of a monolithic structure, and fourth step **3308** and sixth step **3313** are performed simultaneously.

(123) FIG. **34** shows a rear perspective view of an alternate climbing stick **3400** according to another embodiment. Climbing stick **3400** is substantially similar to climbing stick **106**, except that climbing stick **3400** includes an alternate set of feet **3402**. Therefore, the features and elements of climbing stick **3400** that are substantially similar to those of climbing stick **106** are denoted by like reference numbers and not described again in detail the following description.

(124) Climbing stick **3400** includes a set of step structures **2100**, a post **2104**, a set of end caps **2106**, and a set of feet **3402** coupled to a frame **2108**. Each of feet **3402** is a tree engaging structure configured to bite into the side of a tree when climbing stick **106** is strapped to the tree and weight is exerted on at least one of step structures **2100**. As will be described with reference to upcoming FIGS. **35-37B**, feet **3402** provide climbing stick **3400** significantly improved stability as compared to climbing stick **106**.

(125) FIG. **35** shows a rear perspective view of the bottom surface of foot **3402**. Foot **3402** is substantially similar to foot **2102**, except that foot **3402** further includes a ramped surface **3500** formed at the tree-facing surface of each respective one of teeth **2400**. Therefore, the features and elements of feet **2102** that are substantially similar to those of feet **3402** are denoted by like reference numbers and are not described again in detail to avoid redundancy. Accordingly, each of feet **3402** includes an aperture **2122**, a channel **2402**, apertures **2404**, a channel **2406**, and a plurality of antler-shaped teeth **2400** each having a respective ramped surface **3500** formed thereon.

(126) Feet **3402** are formed by first extruding aluminum through an antler-shaped die to form an aluminum beam having an antler-shaped cross-sectional profile. Then, the extruded beam is cut into multiple sections wherein each is used to form a discrete, monolithic one of feet **3402**. Next, a ramp **3500** is machine cut into each respective one of teeth **2400**. Finally, each of feet **3402** undergoes a surface finishing process such as, for example, anodizing, painting, etc. As an alternative to extrusion, each of feet **3402** may be cut (e.g., water-jet cutting, machining, flame cutting, etc.) from a solid pre-strengthened/pre-worked/pre-toughened plate of aluminum (i.e. aluminum alloy) or the like before being subjected to surface finishing. Indeed, the ramped surfaces can even provide the same benefits in tree engaging structures that are cast or formed by any other means.

(127) FIG. **36** shows a side view of one of feet **3402**, which includes a facing surface **3600** having a top edge **3602**, a bottom edge **3604**, and ramped surface **3500** formed therebetween. When

climbing stick **3400** is attached to a tree, top edge **3602** is disposed closer to the center of the tree than bottom edge **3604**. In the example embodiment, the vertical height of ramped surface **3500** is 0.20 inches and ramped surface **3500** is cut at a 45 degree angle as shown. Although only one of the eight teeth **2400** is visible in FIG. **36**, each one of teeth **2400** of foot **3402** includes a facing surface **3600** having a top edge **3602**, a bottom edge **3604**, and ramped surface **3500** formed therebetween.

(128) FIGS. **37A** and **37B** show side views of climbing stick **3400** attached to tree **102** when a small downward force f_a is exerted on step(s) **2100** and a larger downward force F_a is exerted on step(s) **2100**, respectively. When a small force f_a is exerted on steps **2100**, climbing stick **3400** slides slightly down tree **102** while strap **108** remains frictionally fixed to the outer surface of tree **102**. As a result, the tension in strap **108** increases thereby causing an inward horizontal force f_i to be imparted on climbing stick **3400**. This inward horizontal force f_i causes teeth **2400** to urge into the bark of tree **102**, thereby stabilizing climbing stick **3400**. Of course, a larger force F_a onto step **2100** (shown in FIG. **37B**) causes climbing stick **3400** to slide even further down tree **102**, thereby resulting in an even greater inward force F_i . Ramped surfaces **3500** allow climbing stick **3400** to slide even further down tree **102**. As a result, ramped surfaces **3500** cause an even greater inward horizontal force on climbing stick **3400**, thereby increasing the stability of climbing stick **3400**.

(129) The top edges of teeth **2400** have no slope or ramped surface. As a result, the tension in strap **108** causes teeth **2400** to dig into the bark of tree **102** and remain stable, rather than to cause climbing stick **3400** to advance back up tree **102** when the force is removed from steps **2100**. The combination of the sloped bottom surface and flat top surface of teeth **2400** causes a ratchet-like effect when force is exerted onto steps **2100**. That is, climbing stick **3400** may advance down tree **102** but resists moving back upward unless strap **108** is removed. Moreover, the downward motion is smooth, as opposed to the unpredictable motion of prior climbing sticks. Another advantage provided by ramped surfaces **3500** is that feet **3402** have a higher probability of sliding completely past small irregularities (i.e., knots, branches, burls, etc.) in tree **102** that could otherwise engage portions of feet **3402** and cause them to bend or twist when force is exerted on climbing stick **3400**.

(130) FIG. **38** shows a rear perspective view of a bottom surface of an alternate tree engaging structure **3800**, which can be substituted for tree engaging structure **310**. That is, tree engaging structure **310** can be removed from treestand **104** and replaced with tree engaging structure **3800**. The only difference between tree engaging structure **3800** and **310** is that tree engaging structure **3800** includes ramped surfaces **3802**, which provide the same function that ramped surfaces **3500** provide to feet **3402**. The features and elements of tree engaging structure **3800** that are substantially identical to those of tree engaging structure **310** are denoted by like reference numbers and withheld from the following description.

(131) Tree engaging structure **3800** includes a channel **1000**, an aperture **1002**, a set of outside teeth **1004**, a set of middle teeth **1006**, a set of inside teeth **1008**, and a ramped surface **3802** formed on each respective one of teeth **1004**, **1006**, and **1008**.

(132) FIG. **39** shows a bottom plan view of tree engaging structure **3800**. Tree engaging structure **3800** is an aluminum structure formed, for example, by first extruding the antler-shaped profile along an axis into a beam and then cutting the beam perpendicular to the axis. Accordingly, a single beam yields multiple tree engaging structures **3900**. The individual tree engaging structures **3900** then undergo additional machining processes to form ramped surfaces **3802**. As an alternative, tree engaging structure **3800** may be cut from a pre-strengthened plate using a plate-cutting process such as machining, water jetting, flame cutting, etc. Such plate cutting processes may be done manually and/or automatically using CNC systems. In addition, as explained above, the ramped surfaces can also provide the described advantages in tree engaging structures made by casting or any other manufacturing process.

(133) FIG. **40** shows a side view of another example treestand **4000** attached to tree **102** by straps **108**. Treestand **4000** is substantially similar to treestand **104**, except that treestand **4000** includes

tree engaging structure **3800** instead of tree engaging structure **310**, and platform **300** additionally includes integral ramped surfaces **4002**. Although only one of teeth **416** is shown in FIG. **40**, all of teeth **416** include a respective ramped surface **4002** in this example embodiment. Ramped surfaces **3802** and **4002** function similarly and provide the same advantages to treestand **4000** as ramped surfaces **3500** provide to climbing sticks **3400**. That is, ramped surfaces **3802** and **4002** allow treestand **4000** to slide downward over tree irregularities (e.g., knots, burls, branches, etc.) that could otherwise concentrate damaging stresses on tree engaging components of treestand **4000**. Ramped surfaces **3802** and **4002** also allow treestand **4000** to slide smoothly down tree **102**, but the top edges of tree engaging structure **3800** and platform **300** also resist the sliding of treestand **4000** back up tree **102** due to the tension in straps **108**. As previously mentioned, the further the tree mountable structure (e.g., treestand, climbing stick, etc.) slides down the tree, the greater the tension in straps **108** becomes and, therefore, the higher the inward force that pulls the tree mountable structure toward the center of the tree. Further, the greater the force pulling the tree mountable structure toward the center of the tree, the more stable the tree mountable structure becomes.

(134) FIGS. **41A-41E** show various alternate ramped surface shapes that may be substituted for ramped surfaces **3500** of foot **3402**. Furthermore, any of the ramped surfaces shown in FIGS. **41A-41E** may also be substituted for ramped surfaces **3802** and/or **4002**.

(135) FIG. **41A** shows a side view of foot **3402** wherein a linear/planar ramped surface **4100** of foot **3402** extends completely from the top edge of foot **3402** to the bottom edge of foot **3402**.

(136) FIG. **41B** shows a side view of foot **3402** wherein a rounded/curved ramped surface **4102** of foot **3402** extends completely from the top surface of foot **3402** to the bottom surface of foot **3402**. In each of the examples of FIGS. **41A** and **41B**, the apex of the tooth **2400** extends from the top surface of foot **3402** to the bottom surface of foot **3402**.

(137) FIG. **41C** shows a side view of foot **3402** wherein a rounded/curved ramped surface **4104** of foot **3402** extends from a bottom of edge of the apex **4106** of tooth **2400** to the bottom surface of foot **3402**.

(138) FIG. **41D** shows a side view of foot **3402**, wherein a linear/planar ramped surface **4108** of foot **3402** extends from the bottom edge of a straight front facing apex surface **4110** of tooth **2400** to the bottom surface of foot **3402**. Furthermore, a second linear/planar ramped surface **4112** extends downward from the top edge of flat front facing surface **4110** to the top surface of foot **3402**.

(139) FIG. **41E** shows a side view of foot **3402** wherein a linear/planar ramped surface **4114** of foot **3402** extends from a point **4116** to the bottom surface of foot **3402**. A second linear/planar ramped surface **4118** extends downward from point **4116** to the top surface of foot **3402**. Ramped surfaces **4112** of FIG. **41D** and **4118** of FIG. **41E** each increase the resistance to upward movement on the tree.

(140) The description of particular embodiments of the present invention is now complete. Many of the described features may be substituted, altered or omitted without departing from the scope of the invention. For example, alternate treestand and/or climbing stick mounting straps (e.g. rope), may be substituted for the cam straps. As another example, alternate material types (e.g., plastic and/or other polymers, titanium and/or other metals/alloys, etc.), may be substituted for the 6061 aluminum plates from which the example platform and/or seat are/is formed. These and other deviations from the particular embodiments shown will be apparent to those skilled in the art, particularly in view of the foregoing disclosure.

Claims

1. A tree climbing stick comprising: a frame having a top portion, an opposite bottom portion, a front facing surface extending from said top portion to said bottom portion, a rear facing surface

opposite said front facing surface, said rear facing surface extending from said top portion to said bottom portion of said frame; a first tree engaging structure coupled to said front facing surface of said frame at said top portion of said frame in a fixed position with respect to said frame; a second tree engaging structure coupled to said front facing surface of said frame at said bottom portion of said frame in a fixed position with respect to said frame; a first step coupled to said rear facing surface of said frame at said top portion of said frame; a second step coupled to said rear facing surface of said frame at said bottom portion of said frame; and an adjustable fastener configured to attach to said frame to a tree; and wherein said frame is a first, discrete monolithic structure; said first tree engaging structure is a second, discrete monolithic structure cut from a single piece of solid material; said second tree engaging structure is a third, discrete monolithic structure cut from a single piece of solid material; said first step is a fourth, discrete monolithic structure cut from a single piece of solid material and including a first step portion disposed on a first side of said frame and a second step portion disposed on a second side of said frame opposite said first side of said frame; and said second step is a fifth, discrete monolithic structure cut from a single piece of solid material and including a first step portion disposed on a first side of said frame and a second step portion disposed on a second side of said frame opposite said first side of said frame; said first step includes a front facing surface; said front facing surface of said first step defines a channel configured to receive said rear facing surface of said top portion of said frame; said second step includes a front facing surface; said front facing surface of said second step defines a channel configured to receive said rear facing surface of said bottom portion of said frame; said rear facing surface of said frame is seated in said channel of said first step to prevent rotation of said first step with respect to said frame; and said rear facing surface of said frame is seated in said channel of said second step to prevent rotation of said second step with respect to said frame.

2. The tree climbing stick of claim 1, wherein: said first tree engaging structure includes a front facing surface and an opposite rear facing surface; said front facing surface of said first tree engaging structure is configured to engage a tree; said rear facing surface of said first tree engaging structure defines a first channel configured to receive said front facing surface of said top portion of said frame; said second tree engaging structure includes a front facing surface and an opposite rear facing surface; said front facing surface of said second tree engaging structure is configured to engage a tree; said rear facing surface of said second tree engaging structure defines a second channel configured to receive said front facing surface of said bottom portion of said frame; said front facing surface of said of said top portion frame is seated in said first channel of said first tree engaging structure to prevent rotation of said first tree engaging structure with respect to said frame; and said front facing surface of said bottom portion of said frame is seated in said second channel of said second tree engaging structure to prevent rotation of said second tree engaging structure with respect to said frame.

3. The tree climbing stick of claim 2, further comprising a first fastener and a second fastener, and wherein: said first channel and said second channel extend along a first direction; said first tree engaging structure defines an aperture having a first axis perpendicular to said first direction; said top portion of said frame defines a first aperture aligned along said first axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; said first fastener is disposed through said aperture of said first tree engaging structure and said first aperture of said top portion of said frame; said second tree engaging structure defines an aperture having a second axis perpendicular to said first direction and parallel to said first axis; said bottom portion of said frame defines a second aperture aligned along said second axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; and said second fastener is disposed through said aperture of said second tree engaging structure and said second aperture of said bottom portion of said frame.

4. The tree climbing stick of claim 1, further comprising a first fastener and a second fastener, and wherein: said channel of said first step and said channel of said second step extend along a first

direction; said first step defines an aperture having a first axis perpendicular to said first direction; said top portion of said frame defines a first aperture of said frame aligned along said first axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; said first fastener is disposed through said aperture of said first step and said first aperture of said frame; said second step defines an aperture having a second axis parallel to said first axis and perpendicular to said first direction; said bottom portion of said frame defines a second aperture of said frame aligned along said second axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; and said second fastener is disposed through said aperture of said second step and said second aperture of said bottom portion of said frame.

5. The tree climbing stick of claim 1, further comprising a first fastener and a second fastener, and wherein: said channel of said first step and said channel of said second step extend along a first direction; said first step defines an aperture having a first axis perpendicular to said first direction; said top portion of said frame defines a first aperture of said frame aligned along said first axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; said first tree engaging structure defines an aperture aligned along said first axis; said first fastener is disposed through said aperture of said first step, said aperture of said first tree engaging structure, and said first aperture of said top portion of said frame; said second step defines an aperture having a second axis parallel to said first axis; said bottom portion of said frame defines a second aperture of said frame aligned along said second axis and passing completely through said front facing surface of said frame and said rear facing surface of said frame; said second tree engaging structure defines an aperture aligned along said second axis; and said second fastener is disposed through said aperture of said second step, said aperture of said tree engaging structure, and said second aperture of said bottom portion of said frame.

6. The tree climbing stick of claim 5, wherein said frame includes a substantially rectangular cross-section.

7. The tree climbing stick of claim 6, wherein said frame is a tubular structure having a substantially square cross-section.

8. The tree climbing stick of claim 1, wherein: said first tree engaging structure includes a front facing surface and an opposite rear facing surface; said front facing surface of said first tree engaging structure defines a plurality of teeth configured to engage a tree and defines a channel configured to removably cradle a portion of another climbing stick; said rear facing surface of said first tree engaging structure defines a channel configured to receive said front facing surface of said top portion of said frame; said front facing surface of said top portion of said frame is seated in said channel defined by said rear facing surface of said first tree engaging structure to prevent rotation of said tree engaging structure with respect to said frame; said second tree engaging structure includes a front facing surface and an opposite rear facing surface; said front facing surface of said second tree engaging structure defines a plurality of teeth configured to engage a tree and defines a channel configured to removably cradle a second portion of said another climbing stick; said rear facing surface of said second tree engaging structure defines a channel configured to receive said front facing surface of said bottom portion of said frame; and said front facing surface of said bottom portion of said frame is seated in said channel defined by said rear facing surface of said second tree engaging structure to prevent rotation of said second tree engaging structure with respect to said frame.

9. The tree climbing stick of claim 8, wherein: said channel defined by said front facing surface of said first tree engaging structure extends along a first axis; said channel defined by said rear facing surface of said first tree engaging structure extends along a second axis parallel to said first axis; said channel defined by said front facing surface of said second tree engaging structure extends along said first axis; and said channel defined by said rear facing surface of said second tree engaging structure extends along said second axis.

10. The tree climbing stick of claim 1, wherein: said first step includes a first standing surface on said first step portion and a second standing surface on said second step portion, said frame being disposed between said first standing surface of said first step and said second standing surface of said first step, and said second step includes a first standing surface on said first step portion and a second standing surface on said second step portion, said frame being disposed between said first standing surface of said second step and said second standing surface of said second step.
 11. The tree climbing stick of claim 1, wherein said first tree engaging structure includes a front facing surface and an opposite rear facing surface, said front facing surface of said first tree engaging structure is configured to engage a tree, said rear facing surface of said first tree engaging structure defines a protrusion abutting a first side surface of said frame to prevent rotation of said first tree engaging structure with respect to said frame.
 12. The tree climbing stick of claim 11, wherein said rear facing surface of said first tree engaging structure further includes a second protrusion abutting a second side surface of said frame opposite said first side surface of said frame to further prevent rotation of said first tree engaging structure with respect to said frame.
 13. The tree climbing stick of claim 12, wherein: said first protrusion and said second protrusion define opposing sidewalls of a channel, and said front facing surface of said frame is seated in said channel.
 14. The tree climbing stick of claim 1, wherein a portion of at least one of said first and second steps is configured to receive a respective tree engaging structure of another tree climbing stick and to restrain motion of said second tree climbing stick relative to said tree climbing stick along a direction parallel to a longitudinal axis of said frame.
 15. The tree climbing stick of claim 1, wherein: said first step is directly connected to said frame; and said second step is directly connected to said frame.
 16. The tree climbing stick of claim 5, wherein: said frame includes a tubular structure; said first aperture defined by said top portion of said frame includes a first opening through said front facing surface of said frame and a second opening through said rear facing surface of said frame; and said second aperture defined by said bottom portion of said frame includes a third opening through said front facing surface of said frame and a fourth opening through said rear facing surface of said frame.
 17. The tree climbing stick of claim 4, wherein: said frame includes a tubular structure; said first aperture of said frame includes a first opening through said front facing surface of said frame and a second opening through said rear facing surface of said frame; and said second aperture of said frame includes a third opening through said front facing surface of said frame and a fourth opening through said rear facing surface of said frame.
 18. The tree climbing stick of claim 5, wherein: said frame includes a tubular structure; said first aperture of said frame includes a first opening through said front facing surface of said frame and a second opening through said rear facing surface of said frame; and said second aperture of said frame includes a third opening through said front facing surface of said frame and a fourth opening through said rear facing surface of said frame.
 19. The tree climbing stick of claim 1, wherein said frame, said first tree engaging structure, said second tree engaging structure, said first step, and said second step are all integral parts of a single rigid body structure.
 20. The tree climbing stick of claim 1, wherein said frame defines an array of weight reduction apertures between said top portion of said frame to said bottom portion of said frame.
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